# Analyzing Fixation Targets on Real and AI-Generated Faces


**Eemeli Ahokas**
School of Computing
University of Eastern Finland
Joensuu Finland
eahokas@uef.fi

**Muhammad Ilyas**
School of Computing
University of Eastern Finland
Joensuu Finland
milyas@uef.fi

**Joni Kuronen**
School of Computing
University of Eastern Finland
Kuopio Finland.
joniku@uef.fi

**Patrik Laamanen**
School of Computing
University of Eastern Finland
Kuopio Finland
patrilaa@uef.fi

**Meeri-Maria Lakka**
School of Computing
University of Eastern Finland
Kuopio Finland
meeriml@uef.fi

**Petteri Puustinen**
School of Computing
University of Eastern Finland
Joensuu Finland
pettepuu@uef.fi

**Ossi Rytkönen**
School of Computing
University of Eastern Finland
Kuopio Finland
ossir@uef.fi

**Aatu Tolvanen**
School of Computing
University of Eastern Finland
Joensuu Finland
aatt@uef.fi



**ABSTRACT**

The proliferation of artificial intelligence (AI) has introduced transformative possibilities in healthcare and education, yet it also raises concerns, particularly with the surge in deepfake technology. Distinguishing AI-generated faces from real ones remains a challenge, especially these days. In this study, we investigate the facial features capturing human attention during this discrimination process. Four computer science students participated, viewing a set of 10 images—comprising both AI-generated and real faces—while their eye movements were tracked using Tobii Eye Tracker 5. Our quantitative approach aims to understand fixation patterns and how the participants try to figure out which photo is real, and which one is AI-generated. This research contributes to comprehending the cognitive processes involved in differentiating between AI-generated and real faces, providing insights for the development of more effective detection methods and safeguards against malicious AI use. The results of this study suggest that participants fixate on photo features in the following order: skin structure, eyes, hair, mouth, nose, ears, and background. This gives us some insight on what human beings pay attention to while trying to find out if the image is real or AI-generated.

## CCS CONCEPTS

• Human-centered computing → Human computer interaction (HCI) → Empirical studies in HCI • Applied computing → social and behavioral sciences → Psychology • Computing methodologies → Computer graphics → Image manipulation • Computing methodologies → Computer graphics → Graphics systems and interfaces → Perception


## KEYWORDS

Eye tracking, Fixation detection, Image recognition, Visual attention, HCI

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## 1 Introduction

The last decade has seen the popularity of artificial intelligence grow at a remarkable pace. It can be used to aid humans in an unimaginable number of ways. Two concrete examples of fields that can benefit from the use of AI are healthcare and education, where predictive models and pattern recognition can be used to identify health risks and students at risk of failing courses, respectively [1][2]. However, while AI brings great possibilities, it can also be used for malicious purposes. For instance, deepfake generation has been gaining traction in recent years due to the advancements in deep learning. It can be used to influence people politically, taint people's reputation and invade people's privacy. [3]

Being able to detect faces that have been influenced by artificial intelligence is fast becoming an important skill. However, previous research has shown that distinguishing AI-generated faces from real ones is non-trivial, and even nearly impossible. Nightingale and Farid [4] found that good quality synthetic faces are in fact indistinguishable from real ones and are often even perceived as more trustworthy than real faces. [5] and [6] also found similar results: AI-generated images have a high probability of deceiving human beings. Thus, modern AIs can clearly generate convincing faces, at least when pictures with little to no artefacts are picked. Without the presence of obvious artefacts, people have to look hard for clues that might suggest that the person in the picture is not real.

In this paper, we aim to find out what features on people's faces humans tend to pay attention to when they are trying to figure out whether a photo has been AI generated or not. There were 4 participants in our study. All participants are computer science students from University of Eastern Finland. The participants were tested with a series of 10 pictures of human faces. The images included 6 AI-generated images and 4 real images. Participants did not know how many real and how many AI-generated images were in the set. Each image was shown for 10 seconds. During these 10 seconds, the participants had to decide in their mind whether the image was an AI generated or a real face image. The AI-generated images were downloaded from www.thispersondoesnotexist.com, which is a free website that generates human faces. Images that had no obvious rendering artifacts were chosen so that the participants would have to look for smaller clues instead of fixating on just one obvious spot. The real human photos were downloaded from www.pexels.com, which is a website containing copyright free images.

We collected the eye movement data with Tobii Eye Tracker 5 and used Tobii Recorder software to collect the data into a csv-file from each participant. Before collecting the data, we calibrated the eye-tracking device by administering Tobii 5 software's own calibration for each participant. Following the calibration, we asked each participant to complete the task. The experiment was conducted on a screen with a width of 1536 pixels and height of 864 pixels. After the data was collected the data was preprocessed by removing all the csv rows where the participant blinked or looked outside of the screen. Some rows were also removed corresponding to the short times before and after the experiment.

There might be errors in the data due to the device not being mounted in the exact center position of the screen (0.5 -1.0 cm off), and not cleaning the tracker lens thoroughly before using it. One participant was wearing eyeglasses which can also affect the data. Involuntary head movements can also introduce errors by altering the perceived trajectory of the eyes, affecting the accuracy of gaze tracking.

We hypothesize that the features the participants fixate on the most are skin structure, eyes, ears, hair, background, mouth, and nose, in that order, measured by fixation duration on those features. This hypothesis is based on results found by Yegemberdiyeva and Amirgaliyev [6]. As part of their study, they documented how often participants relied on specific photo features to identify whether photos were AI-generated. However, it is important to note their study results were collected qualitatively. This differs from our study, where the data is gathered strictly through eye tracking. Their study listed the quality of picture, the combination of all listed features, and the symmetry and proportion of the face as features, but we are not able to reliably identify those from our eye tracking data. Thus, we are not including them in our study.

## 2 The algorithm

The I-DT algorithm is used for fixation detection, a key part of studying eye movement. It works by checking two things: how much the eyes move (dispersion) and how long they stay in one place (duration). The algorithm looks at eye tracking data point by point. If the points do not move much and stay in the same area for a certain time, it counts as a "fixation" - a moment when the eyes are steady, looking at one spot.

To do this, the algorithm uses a set of rules. It looks at a group of data points and measures how far apart they are. If they are close together and stay like that for long enough, it marks this as a fixation. Then, it moves to the next group of points and does the same thing. This way, it can tell us when and where someone was focusing their gaze as they looked at something.

## 2.1 Algorithm Settings

The algorithm was tested using data from four participants, each observing ten different images. For each image, the algorithm was applied twice, using two distinct sets of thresholds. We selected these settings based on trials with multiple different settings. These settings provided the most realistic number of fixations out of the trialed settings.

Setting 1:
Dispersion Threshold: 35 units
Duration Threshold: 10 units
Setting 2:
Dispersion Threshold: 60 units
Duration Threshold: 20 units
Procedure:
Each participant's eye-tracking data, consisting of x and y coordinates over time, were analyzed using the above settings. The algorithm's task was to classify segments of the data as either fixations or saccades (rapid eye movements) based on the defined thresholds.

## 2.2 Algorithm pseudocode

```
Algorithm: I-DT (raw_data, dispersion_threshold, duration_threshold)
1. Initialize an empty list for fixations.
2. Set the initial index to 0.
3. While (initial index + duration_threshold) <= length of raw data:
  a. Set window to raw_data from initial index to initial index + duration_threshold.
  b. Calculate dispersion for the window.
  c. If dispersion <= dispersion_threshold:
    i. Set start index to initial index.
    ii. Set the end index to the last index of the window.
    iii. While (dispersion <= dispersion_threshold AND end index < length of raw data):
        - Increment end index by 1.
        - Add the next sample to the window.
        - Recalculate dispersion for the window.
    iv. Set end index to the last index before dispersion exceeded dispersion_threshold.
    v. Calculate centroid for the fixation using samples from start index to end index.
    vi. Record fixation with start time, end time, centroid, and duration.
    vii. Set initial index to end index + 1.
  d. Else:
    i. Increment the initial index by 1.
4. Return the list of fixations.

Function: Calculate Dispersion (window)
1. Set x_coords as a list of all the x points of window.
2. Set y_coords as a list of all the y points of window.
3. Calculate dispersion as (max(x_coords) – min(x_coords)) + (max(y_coords) – min(y_coords)).
4. Return dispersion.

Function: Calculate Centroid (window)
1. Set x_coords as a list of all the x points of window.
2. Set y_coords as a list of all the y points of window.
3. Calculate centroid_x as sum(x_coords) / len(x_coords).
4. Calculate centroid_y as sum(y_coords) / len(y_coords).
5. Return (centroid_x, centroid_y).
```

## 2.3 Link to notebook

Link to Google Collab: https://colab.research.google.com/drive/1c2eZz6l7Jz-K343UFW4b30eCIPdi765x#scrollTo=HYpK8wVZvJn2

## 2.4 Output CSV Snippet

| Subject_ID | MFD_Eyes (ms) | MFD_nose (ms) | MFD_SD_eys (ms) | MFD_SD_nose (ms) | MFD_Overall (ms) | MFD_SD_Overall (ms) |
|---|---|---|---|---|---|---|
| Participant_1 | 407.7 | 343.4 | 126.2 | 14.3 | 473.8 | 231.3 |
| Participant_2 | 333.3 | 472.2 | 30.3 | 228.3 | 407.0 | 132.5 |
| Participant_3 | 571.8 | 643.9 | 281.1 | 390.1 | 499.2 | 257.4 |
| Participant_4 | 492.4 | 387.9 | 269.7 | 92.7 | 472.3 | 229.4 |

**Table 1: Snippet of the mean fixation duration csv file output by the code.**

# 3 Data analysis

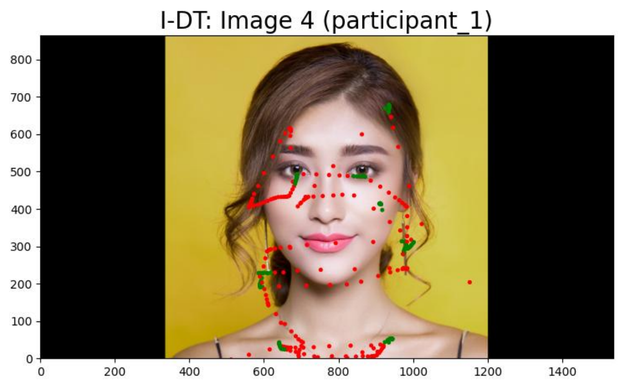


**Image 1. Scatterplot of the fixations and saccades**

Image 1 shows scatter plot of our eye-tracking experiment. Red dots and lines across the person's face indicate where the participant's gaze paused (fixations) and moved swiftly (saccades). The clustering of red dots around her eyes, lips, lower neck, and the contour of her face suggest these areas were of particular interest to the observer. But at the top of her head, where her hair is, there are no dots. This shows that the person did not look at her hair much. This picture helps us understand what parts of a face people look at when they are trying to tell if it is a real person, or a picture made by a computer.

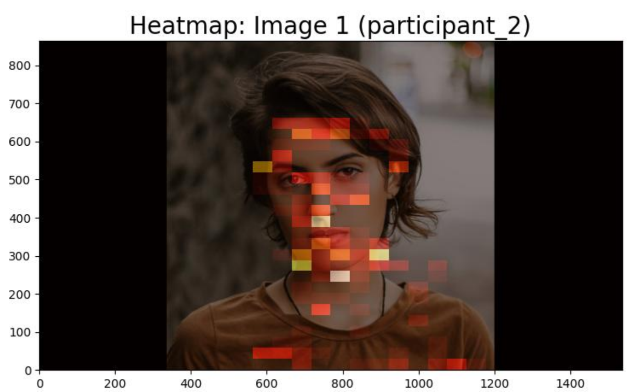


**Image 2. Heatmap of the fixations**

The image 2 is a heatmap on a person's face from our eye-tracking experiment, showing where someone looked most. The red and orange spots show a lot of attention on the eyes and nose. This tells us these features were the most interesting to the viewer.

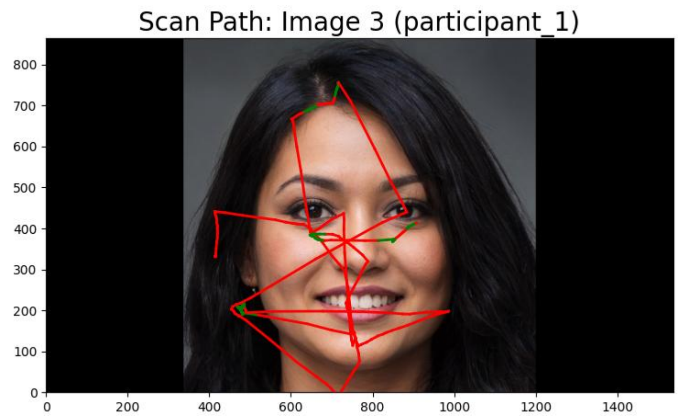


**Image 3. Scan path**

Image 3 shows the scan path of the gaze. The lines crisscross mainly over the person's eyes and mouth, which were the focus points. This tells us how the participant's eyes moved while looking at the face. This behavior is similar to previous images.

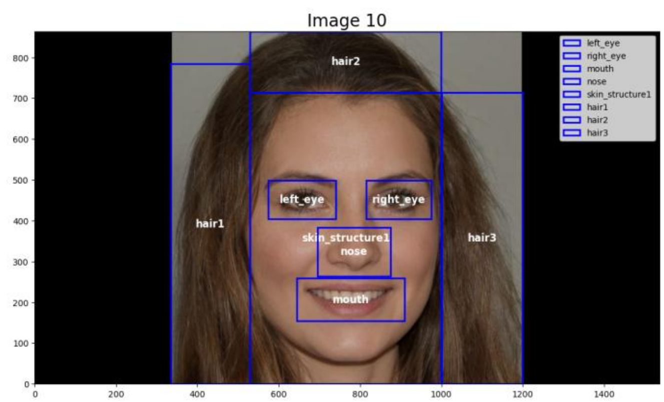


**Image 4. Image showing the areas of interest in the image 10**

| Feature | Number of fixations | Percentage of fixations | Total fixation duration (s) | Mean fixation duration (ms) | Std for fixation duration (ms) |
|---|---|---|---|---|---|
| Skin structure | 128 | 32.08 % | 56.18 | 439.92 | 179.67 |
| Eyes | 87 | 21.8 % | 44.60 | 512.71 | 265.50 |
| Ears | 23 | 5.8% | 11.27 | 490.11 | 252.06 |
| Mouth | 55 | 13.8 % | 24.72 | 449.58 | 206.96 |
| Nose | 29 | 7.3 % | 13.15 | 453.50 | 231.12 |
| Hair | 56 | 14.03 % | 26.84 | 479.92 | 219.87 |
| Background | 21 | 5.26 % | 9.24 | 440.11 | 253.29 |
| Total | 399 | 100 % | 186 | | |

**Table 2: Comprehensive summary statistics for participants and images**

The collected data was filtered, we removed all invalid data from the csv-file. We also removed all the data before the images were shown to participants and after the image slideshow ended.

To get these results we tested two settings on the algorithm, this is discussed in chapter 2. From the algorithm we used we gathered the data shown in table 2. We combined all the data from all the participants and the images and grouped them by features. In total there were 399 fixations. The mean fixation duration per feature and the total duration can be seen from table 2. Table 2 shows that the most attention was paid to skin structure, eyes, hair, mouth, nose, ears, and background, in that order, when we look at the total fixation duration column.

## 4 Discussion on results

In this study we tried to find out what features on people's faces humans tend to pay attention to when they are trying to figure out whether a photo has been AI generated or not. Our hypothesis was that the participants in this experiment fixate on the most are skin structure, eyes, ears, hair, background, mouth, and nose, in that order, measured by fixation duration on those features. In our study the most attention was paid to skin structure, eyes, hair, mouth, nose, ears, and background, in that order, so there is some difference to the hypothesis.

Like the hypothesis, in our study participants had the greatest number of fixations on the skin structure as seen on table 2. The second most fixations got the eyes. One plausible reason these two are the most fixated features, is because the eyes and facial skin structure play a vital role in conveying emotions and intentions. The reason that ears received the second least attention might be because we had multiple photos where the subjects' ears were not fully visible, for example we had an image where the ears were concealed by hair as seen in image 4.

While background got the least number of fixations from the study participants it might be because the task was to guess if the face in the image was AI-generated or not. If the task would've been to evaluate a different set of images, e.g. people in different environments or full body images, the participants might've paid more attention to the background. This might also be because there were images that did not include background, or the background was just in one color.

It's worth considering the possibility of errors in fixation classification within areas of interest. This is probably caused by our method of plotting coordinates using square boxes. A case in point is Image 4, where a significant portion of the background is inside the "hair boxes," suggesting a potential distortion in the fixation analysis.

Our result may also differ from the hypothesis since we had a small sample size and only 10 images, we tested the participants with. Also limited time per image may cause participants to hurry with scanning the image and therefore may cause difference in the fixations. Participants' possible head movements may have also caused some difference in fixations.

In conclusion, our study shows the overall trend in our study supports the initial hypothesis while there are some deviations in the order of fixation durations for specific features. The participants fixated the most on the skin structure and eyes of the faces in the images when trying to figure out whether the image was real or AI-generated.

5 Members and contributions

Table 3 contains the estimated contributions of each group member to the group homework tasks and course project. Overall, it's fair to say that each member contributed to every task, and there were no massive outliers in terms of amount of contribution. We worked as a group and achieved our goals together.

| **Member** | **Group Homework contribution** | **Project contribution** |
|---|---|---|
| Eemeli Ahokas | 10 % | 12.5 % |
| Muhammad Ilyas | 10 % | 12.5 % |
| Joni Kuronen | 10 % | 12.5 % |
| Patrik Laamanen | 20 % | 12.5 % |
| Meeri-Maria Lakka | 20 % | 12.5 % |
| Petteri Puustinen | 10 % | 12.5 % |
| Ossi Rytkönen | 10 % | 12.5 % |
| Aatu Tolvanen | 10 % | 12.5 % |

**Table 3: The percentage of contribution of each member to group homework tasks and the course project.**

## ACKNOWLEDGMENTS

While we did our best to get scientifically significant results, there are a number of factors that might have influenced our findings. Therefore, the results of this study should be approached with some caution. However, we strongly encourage other researchers to use eye tracking to study this topic, as we feel it has a lot of potential to provide new insights.

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